

Aerospace & Defence, US Telecom & Communications Infrastructure

SpaceX: Potential for MVNO D2D growth? Continuing our assessment of SpaceX Connectivity business and telco impact



Douglas S. Harned, Ph.D.
+1 917 344 8430
douglas.harned@bernsteinsg.com



Stacy A. Rasgon, Ph.D.
+1 213 559 5917
stacy.rasgon@bernsteinsg.com



Laurent Yoon
+1 917 344 8502
laurent.yoon@bernsteinsg.com



Madison Rezaei
+1 917 344 8622
madison.rezaei@bernsteinsg.com



Nestor Wester
+44 20 7676 7067
nestor.wester@bernsteinsg.com



Martin Boruchowicz
+1 917 344 8564
martin.boruchowicz@bernsteinsg.com



Andrew Chung
+1 917 344 8302
andrew.chung@bernsteinsg.com



Nancy Wu
+1 917 344 8545
nancy.wu@bernsteinsg.com

As SpaceX prepares for Starship launch 14, the company plans to deploy the first operational V3 Starlink satellites. In this note, we continue our focus on the Connectivity portion of the business and the approach we see as most attractive for direct-to-device service. We previously discussed the technical challenges of D2D ([SpaceX: The Telecom Play - Can Direct-to-Device Work?](#)) and the hurdles if the company were to build out its own terrestrial network ([SpaceX: More on Direct-to-Device Mobile - what would a buildout cost? Our detailed bottom-up estimates](#)). Here, we assess the potential for an MVNO option, which we have seen as most attractive since our initiation on SpaceX. **Despite historical challenges of MVNOs, we believe the Starlink opportunity could be materially better. In addition, the global reach of Starlink and strength of its broadband offering can create benefits beyond D2D services alone.**

Most important for SpaceX's valuation remains gaining Starship reusability to enable orbital data centers. The company now expects second stage reuse to be demonstrated in Q4. Although less critical, we think this reuse will still be important for Starlink broadband and Starlink Mobile constellations beginning in 2027.

Typical MVNO positioning: a tough business. MVNOs are generally positioned as lower-cost, value-oriented alternatives to the facilities-based players. This positioning leads to lower pricing, while customer acquisition still requires substantial marketing spend, physical retail footprint and, in many market including the US, aggressive device promos.

Without owner economics, these dynamics tend to produce low-margin and often unattractive standalone businesses. As a result, MVNOs typically target specific customer segments where they can differentiate beyond price alone, such as Consumer Cellular's focus on seniors.

The US CableCos provide an interesting development. Comcast and Charter have built meaningful wireless businesses over the past ~8 years, but only through aggressive pricing, free line offers, device promos, and bundled discounts. Importantly, wireless is not the profit center. It serves primarily to protect and enhance the broadband franchise—at least in the foreseeable future. Viewed on a standalone basis, wireless profitability remains modest, and the economics work because the bundle improves overall CLV. That said, meaningful market share gains are possible. Several MVNOs have captured meaningful market share over time, but it typically comes at the expense of profitability.

Why Starlink's MVNO economics could be better. Traditional MVNO economics are constrained by four factors: CAC, device subsidies, lower service pricing, and network access fees paid to MNOs. For Starlink, the first three factors are largely dictated by market structure and competitive dynamics and, therefore, remains inevitable, particularly while the business remains subscale. Network costs, however, may be different.

Continues on the following page...

The average US wireless subscriber offloads roughly 80% of data consumption to Wi-Fi, leaving around 20% over cellular networks. That remaining traffic is generally what drives wholesale network costs for MVNOs. Cable operators have already demonstrated the ability to push offload rates >85% through Wi-Fi first architectures.

Starlink could potentially achieve even higher levels of offload if a meaningful portion of outdoor mobile usage can be carried directly over its satellite network—"D2D"—or perhaps over a femtocell network in certain areas. Significant engineering challenges remain, including seamless handoffs between terrestrial and satellite networks and the limitations inherent in satellite beamforming and capacity management. But if those challenges are addressed over time, Starlink could materially reduce its reliance on wholesale cellular capacity.

The result is an MVNO model with potentially better economics than a traditional MVNO, particularly when bundled with Starlink's high-margin BB products.

MVNO structure could support SpaceX's objectives. We believe a hybrid MVNO-based model represents the faster and most capital-efficient path toward SpaceX's wireless ambitions. Building a nationwide terrestrial network from scratch would require enormous capital and years of execution, whereas an MVNO model offers a faster path to market.

More importantly, wireless could strengthen Starlink's broader BB ecosystem. By bundling mobile and BB services, SpaceX can expand the reach of its much more profitable O&O connectivity platform while improving customer retention and CLV.

But the larger strategic opportunity may extend beyond wireless economics altogether. In a future where AI becomes ubiquitous and the smartphone in your pocket remains the primary edge device, owning the customer relationship may matter more than maximizing wireless margins. Viewed through that lense, the objective is not necessarily to optimize the economics of a standalone wireless business, whether through an MVNO or a facilities-based network. The objective is to achieve scale, establish distribution, and own the customer relationship as quickly as possible. If that is the goal, an MVNO-based hybrid network may be the most practical path forward.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	11 Sep	Price	TTM	Adjusted EPS				Adjusted P/E (x)		
			2026		Rel.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
			Closing	Target	Perf.							
			Price									
SPCX (SpaceX)	O	USD	151.21	248.00	NA	USD		(0.20)	1.30		(765.2)	116.1
T (AT&T)	O	USD	26.06	25.00	(27.9)%	USD	2.12	2.35	2.68	12.3	11.1	9.7
VZ (Verizon)	M	USD	50.61	47.00	(1.6)%	USD	4.68	5.00	5.20	10.8	10.1	9.7
TMUS (T-Mobile)	M	USD	182.33	220.00	(40.7)%	USD	9.72	10.68	13.24	18.8	17.1	13.8
CMCSA (Comcast)	M	USD	25.20	28.00	(35.7)%	USD	4.30	3.40	3.85	5.9	7.4	6.6
CHTR (Charter)	M	USD	145.77	150.00	(61.0)%	USD	36.20	38.98	40.75	4.0	3.7	3.6
AMT (American Tower)	O	USD	177.85	214.00	(25.8)%	USD	5.40	7.01	6.93	32.9	25.4	25.7
CCI (Crown Castle)	O	USD	75.62	96.00	(37.1)%	USD	1.02	2.01	3.22	74.4	37.7	23.5
SBAC (SBA Comm)	M	USD	185.47	219.00	(23.2)%	USD	11.76	8.81	9.57	15.8	21.0	19.4
SPX			7,656.98									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate SpaceX Outperform, with a target of \$248. The case for a multi-trillion dollar valuation continues to be first about delivering on the orbital data center opportunity. Still, we forecast the Connectivity business to generate more than \$100bn in EBITDA by 2031, with more upside possible if the D2D business is successful - which we believe will require a strong MVNO solution. The case for SpaceX was described in our initiation report ([Space Exploration Technologies Corporation: Doing Time in the Universal Mind - Initiate at Outperform; Target Price \\$239](#)) and updated after Q2 reporting ([SpaceX: Solid Q2 with no issues; Projects more upside with higher compute pricing; Starship described on track - Outperform](#)).

We maintain our Outperform rating for T (PT: \$25), and Market-Perform rating for VZ (PT: \$47), TMUS (PT: \$220), CMCSA (PT: \$28), and CHTR (PT: \$150) - all unchanged.

We maintain our Outperform ratings for AMT (PT: \$214) and CCI (PT: \$96) and our Market-Perform rating for SBAC (PT: \$219).

DETAILS

THE GLOBAL WIRELESS MARKET IS MOSTLY SATURATED

In developed markets, network differentiation has become increasingly difficult. Every operator claims to be “#1” on some metric, making meaningful performance advantages harder for consumers to discern. As a result, carriers increasingly compete through brand positioning, customer segmentation, pricing, and bundling rather than network superiority alone.

US Wireless Market is Saturated

From the rise of the internet in the 1990s to the mobile era of the late 2000s, the world underwent a profound digital transformation. Smartphones became ubiquitous, and internet connectivity is now an essential part of daily life. Today, nearly every adult in the U.S. owns a smartphone, with many maintaining separate lines for personal and work use. The introduction of connected devices such as tablets and smartwatches has further increased the prevalence of multiple wireless connections per person.

Smartphone ownership has also become increasingly common among teenagers, while children are obtaining mobile service at younger ages. Together, these trends have pushed the U.S. wireless market toward saturation ([Exhibit 1](#)). Current market penetration has already reached approximately 90%.

Despite operating in a saturated market, the total number of wireless connections has continued to grow at roughly 2% annually. We expect this trend to persist, driven by three fundamental factors: population growth, increases in penetration rates, and greater adoption of multiple connected devices. Higher penetration is primarily supported by rising smartphone adoption among younger age cohorts, while multi-device adoption reflects both the growing use of separate work and personal lines and the increasing penetration of connected devices that require wireless connectivity.

That said, the market's long-term growth potential remains limited. As a result, the emergence of a new large-scale share gainer, such as a Starlink MVNO service, may face challenges. The cable operators provide a useful precedent, as they successfully followed this strategy and gained market share over time.

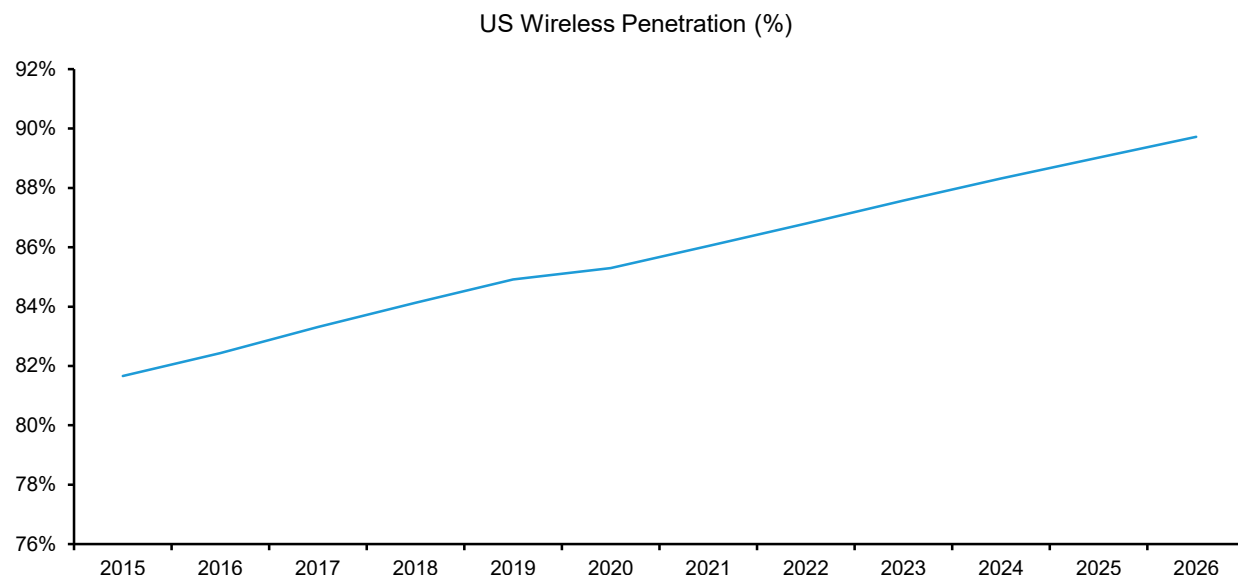
INTERNATIONAL WIRELESS MARKETS ARE LARGELY SATURATED, BUT POCKETS OF UNDERPENETRATION REMAIN

Most international wireless markets also exhibit signs of saturation. As a point of reference, the ratio of total mobile subscriptions to population for several key markets is: the U.S. (133%), Germany (166%), France (144%), Japan (170%), China (125%), and Chile (164%).

However, several large geographic markets, including Canada (106%), Australia (102%), India (82%), and Brazil (102%), appear relatively underpenetrated compared with their peers. These markets may present an opportunity for a provider that can leverage satellite coverage as a complement to a terrestrial MVNO agreement, such as Starlink ([Exhibit 2](#)).

EXHIBIT 1: The US wireless market is already saturated at ~90% penetration. The opportunity for a Starlink MVNO service is in taking share from incumbents

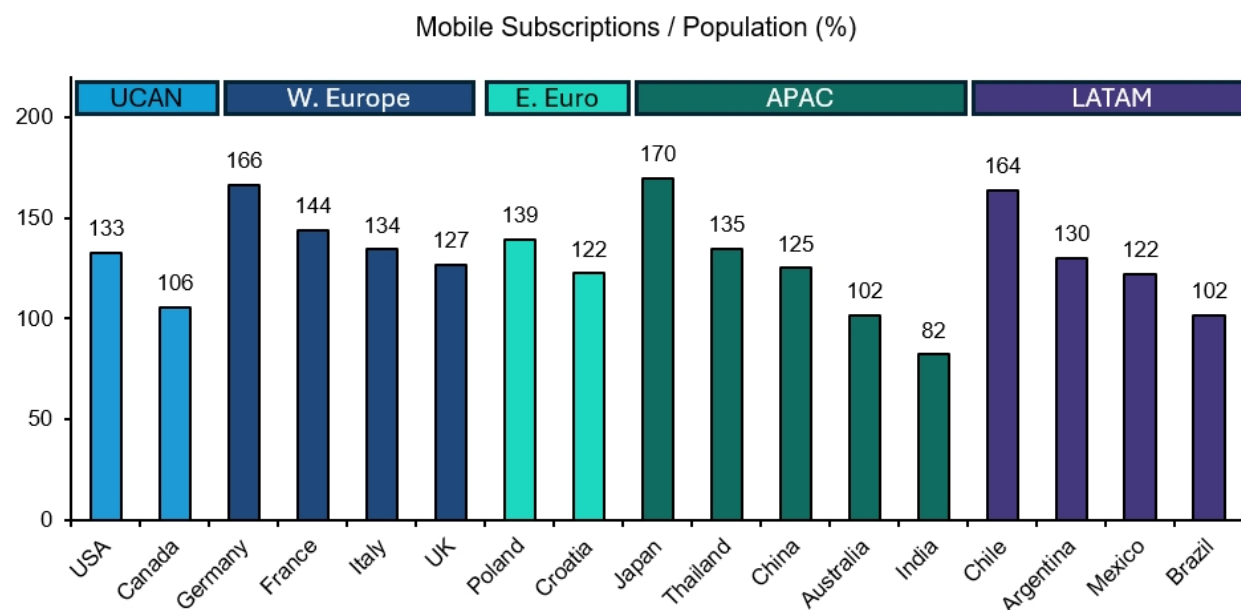
Adjusts for individuals with more than 1 line.



Source: SNL Kagan, Bernstein estimates and analysis

EXHIBIT 2: Most developed and developing markets are saturated or nearing saturation

Data for African markets not available. Does not adjust for individuals with more than 1 line.



Source: SNL Kagan, Bernstein estimates and analysis

THE MVNO ECOSYSTEM

MVNO (Mobile Virtual Network Operator) agreements, in which a Mobile Network Operator (MNO) sells access to its network at wholesale rates, are common across global wireless markets. MVNOs typically market wireless services under their own brands, own the customer relationship, and compensate MNOs based on the network usage generated by their subscribers.

These arrangements are generally attractive to MNOs when the MVNO can acquire customer segments that the carrier would not otherwise be able to reach efficiently.

In the U.S., all three major wireless carriers, Verizon, AT&T, and T-Mobile, participate in MVNO partnerships. Notable independent MVNO relationships include Xfinity Mobile, Spectrum Mobile, and Cox Mobile on Verizon's network; Consumer Cellular, Boost Mobile, and H2O Wireless on AT&T's network; and Optimum Mobile, and Astound Mobile on T-Mobile's network. The model is particularly important for Verizon, which maintains the industry's two largest MVNO agreements with Comcast (Xfinity Mobile) and Charter Communications (Spectrum Mobile) ([Exhibit 3](#)).

For broadband providers such as Comcast, Charter, Optimum, and Cox, entering into MVNO agreements is often attractive because customer acquisition costs can be significantly lower than those of a traditional wireless carrier. By cross-selling mobile services to existing broadband subscribers, these companies can leverage established customer relationships, brand awareness, and billing platforms, reducing both marketing expenses and customer churn.

EXHIBIT 3: Examples of US MVNOs: Verizon has the largest individual agreements through CMCSA (Xfinity) and CHTR (Spectrum)

	Verizon	AT&T	T-Mobile
Major Independent MVNO Relationships	• Xfinity Mobile • Spectrum Mobile • Cox Mobile	• Consumer Cellular • Boost Mobile • H2O Wireless	• Optimum Mobile • Astound Mobile

Source: Company websites, Company filings, Bernstein analysis

CAN MVNOS SCALE?

MVNOs can take meaningful market share

Despite U.S. wireless subscriber growth averaging only about 2% annually, Charter and Comcast have grown their mobile subscriber bases at an approximately 60% CAGR over the past decade through their MVNO strategies ([Exhibit 4](#)). Through their MVNO agreements with Verizon, Charter and Comcast have increased their combined share of the postpaid phone market from zero to approximately 7% in less than a decade. During the same period, Verizon's market share declined from roughly 40% to the low-30% range ([Exhibit 5](#)).

The experience is not unique to the U.S.. In international markets such as the UK and Italy, MVNOs have collectively captured market shares in the high teens. A Starlink MVNO service could potentially expand wireless penetration in certain underserved markets where satellite connectivity complements existing terrestrial networks ([Exhibit 6](#)).

The success of the cable operators' wireless businesses, coupled with the corresponding loss of market share by their host network partner, Verizon, raises an important question: are large MVNO agreements ultimately beneficial for MNOs, or do they primarily cannibalize the carriers' natural addressable customer base? This question becomes particularly relevant if MNOs have to consider a future partnership with Starlink.

MVNOs often rely on aggressive offers to gain share

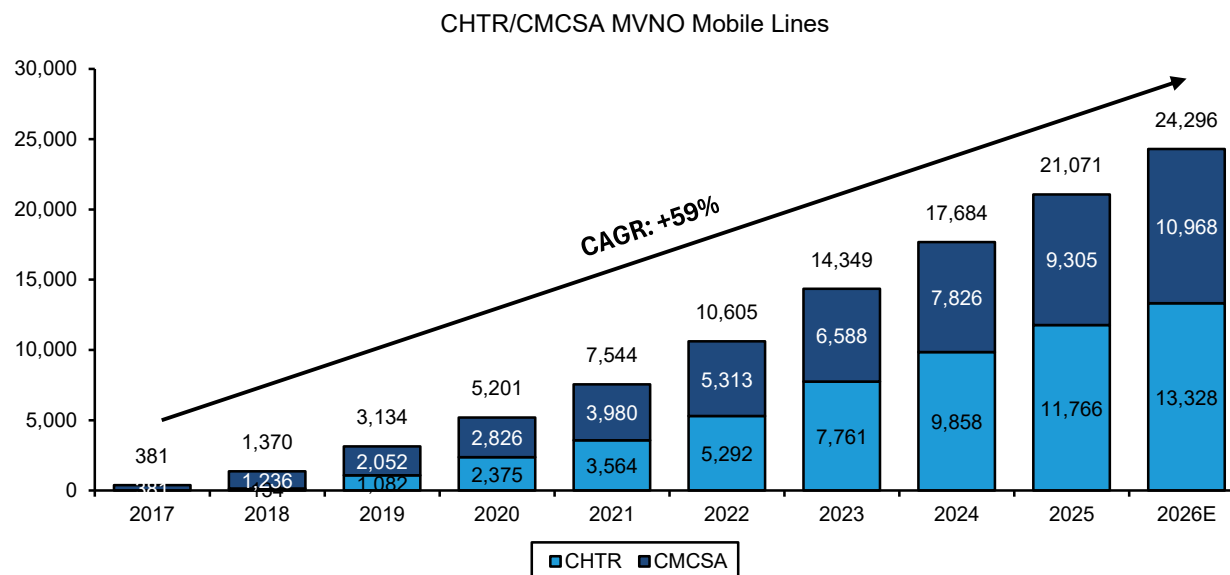
To encourage customers to switch from well-established MNO brands, MVNOs frequently rely on highly competitive promotional offers. These promotions typically include a combination of free service for a period of time, lower monthly rates than the host MNO, and attractive device subsidies.

For example, Xfinity Mobile's value-oriented offering generally provides lower device subsidies than those offered by the major MNOs. However, the service is free during the first year and remains materially cheaper thereafter. As a result, customers can

save approximately \$45 to \$60 per month during the first year of service ([Exhibit 7](#)).

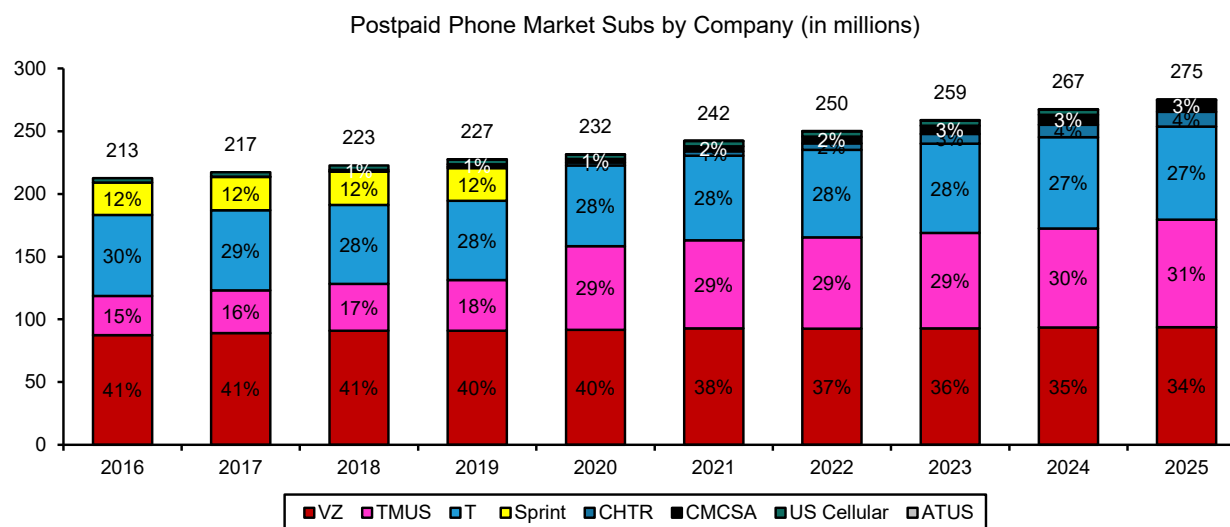
Under Xfinity Mobile's premium plan, customers pay a discounted monthly fee during the first year, followed by a relatively low ongoing rate. In addition, device subsidies are broadly comparable to those offered by the national carriers. Under this plan, customers can save approximately \$95 to \$105 per month during the first year ([Exhibit 8](#)).

EXHIBIT 4: CableMVNOs have grown rapidly in the US market

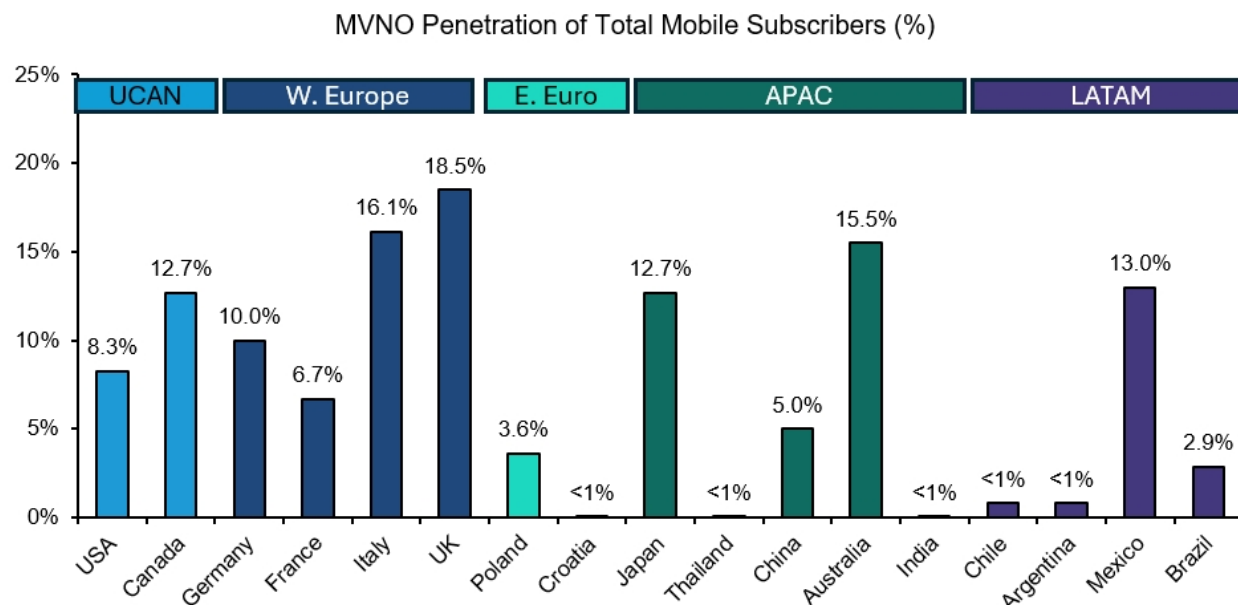


Source: Company Filings, Bloomberg, Bernstein estimates and analysis

EXHIBIT 5: CHTR and CMCSA went from 0% to ~7% of the postpaid phone market subs in less than ten years through MVNO agreements with Verizon



Source: Company Filings, Bloomberg, Bernstein estimates and analysis

EXHIBIT 6: **Some markets have meaningful MVNO market shares**

Source: SNL Kagan, Bernstein estimates and analysis

EXHIBIT 7: **Xfinity's MVNO model offers less device subsidies with its value plan than the MNOs, but the service is free in year 1, and considerably cheaper afterwards**

	New Phone iPhone 17	Trade-in Phone iPhone 16	Xfinity Mobile Plan Mobile Select	Xfinity's value plan
	Xfinity	AT&T	T-Mobile	Verizon
Less device subsidy than competitors	PHONE iPhone 17 \$14.06/mo for 36 mos Includes \$324.00 trade-in credit	PHONE iPhone 17 \$9.16/mo for 36 mos Includes \$500.00 trade-in credit	PHONE iPhone 17 \$2.77/mo for 36 mos Includes \$730.00 trade-in credit	PHONE iPhone 17 \$6.38/mo for 36 mos Includes \$600.00 trade-in credit
Service is free for 1 year, cheaper than comps afterwards	MOBILE PLAN Mobile Select \$99-\$0/mo/1 line Includes \$30/mo credit for 12 months	MOBILE PLAN Value 2.0 \$50/mo/1 line	MOBILE PLAN Essentials 2.0 \$60/mo/1 line	MOBILE PLAN Unlimited Welcome \$65/mo/1 line
	Plan Features ^ • Data: Unlimited high-speed (may slow after 50GB) • Hotspot: 15GB high-speed data, then unlimited at reduced speeds	Plan Features ^ • Data: Unlimited (may slow during congestion) • Hotspot: 3GB high-speed data, then unlimited at reduced speeds	Plan Features ^ • Data: Unlimited (reduced speeds after 50GB) • Hotspot: Unlimited at reduced speeds	Plan Features ^ • Data: Unlimited (may slow during congestion) • Hotspot: Not included
	TOTAL FOR PHONE + MOBILE PLAN \$14.06 /mo Taxes and fees are not included	TOTAL FOR PHONE + MOBILE PLAN \$59.16 /mo Taxes and fees are not included	TOTAL FOR PHONE + MOBILE PLAN \$62.77 /mo Taxes and fees are not included	TOTAL FOR PHONE + MOBILE PLAN \$71.38 /mo Taxes and fees are not included
	Shop This Deal Xfinity Internet Required	Xfinity saves you up to \$45 /mo	Xfinity saves you up to \$49 /mo	Xfinity saves you up to \$57 /mo
		Annual savings decline by \$30 after the 1 st year		

Savings comparison based on 1 line of Xfinity Mobile Select vs 1 line of T-Mobile Essentials, Verizon Unlimited Welcome, and AT&T Extra 2.0 (as of 03/24/26). Taxes & fees extra.

Source: Company website, Bernstein analysis

EXHIBIT 8: In its premium MVNO plan, Xfinity's customers pay a discounted fee on year 1 with a cheaper run rate fee afterwards. Additionally, device subsidies are on par with MNOs

	Xfinity	AT&T	Verizon	T-Mobile
PHONE	iPhone 17 \$0.00/mo for 36 mos Includes \$829.99 trade-in credit Device protection: Included	iPhone 17 \$3.61/mo for 36 mos Includes \$700.00 trade-in credit Device protection: \$19/mo	iPhone 17 \$0.00/mo for 36 mos Includes \$830.00 trade-in credit Device protection: \$19/mo	iPhone 17 \$0.00/mo for 36 mos Includes \$830.00 trade-in credit Device protection: \$19/mo
MOBILE PLAN	Mobile Plus \$45-\$15/mo/1 line Includes \$30/mo credit for 12 months	Premium 2.0 \$109/mo/1 line	Unlimited Ultimate \$114/mo/1 line	Experience Beyond 2.0 \$119/mo/1 line
Plan Features	- Data: Unlimited high-speed - Hotspot: 50GB high-speed data, then unlimited at reduced speeds	- Data: Unlimited high-speed - Hotspot: 100GB high-speed data, then unlimited at reduced speeds	- Data: Unlimited high-speed - Hotspot: 200GB high-speed data, then unlimited at reduced speeds	- Data: Unlimited high-speed - Hotspot: 250GB high-speed data, then unlimited at reduced speeds
TOTAL WITH DEVICE PROTECTION	\$15.00 /mo Taxes and fees are not included	\$112.61 /mo Taxes and fees are not included	\$114.00 /mo Taxes and fees are not included	\$119.00 /mo Taxes and fees are not included
	Shop This Deal Xfinity Internet Required	Xfinity saves you up to \$98 /mo	Xfinity saves you up to \$99 /mo	Xfinity saves you up to \$104 /mo

Annual savings decline by \$30 after the 1st year

Source: Company website, Bernstein analysis

STANDARD MVNO ECONOMICS VS STARLINK MVNO ECONOMICS

Lack of network ownership usually results in modest profitability

One of the most important variables in MVNO economics is the wholesale fee that the MNO charges the MVNO for access to its network. In a simplified framework, the MNO and MVNO agree on a rate per gigabyte (GB) of data consumed. The MVNO's total network expense is then determined by multiplying this rate by the volume of data usage generated by its subscribers. All else being equal, an MVNO's cost structure improves when it can offload traffic away from the host MNO's network. For example, when a subscriber uses home Wi-Fi rather than the terrestrial mobile network, the MVNO avoids incurring wholesale network costs on that usage ([Exhibit 9](#)).

To reduce wholesale expenses, MVNOs seek to maximize traffic offloading. In the U.S., approximately 80-90% of handset data traffic is already offloaded to Wi-Fi as a result of consumer usage patterns, since the most data-intensive activities typically occur at home, in the office, or in other locations with Wi-Fi access. Cable MVNOs such as Comcast and Charter have been able to achieve approximately 87-90% traffic offloading through a combination of extensive Wi-Fi hotspot networks and owned CBRS spectrum network, representing roughly 2-7 percentage points of incremental offloading relative to traditional MNOs. A potential Starlink MVNO offering could leverage its satellite constellation and femtocells to offload traffic in away-from-home use cases. Unlike cable operators' CBRS networks, satellite coverage is largely ubiquitous, potentially enabling even higher levels of traffic offloading ([Exhibit 10](#)).

Because MVNOs typically rely on aggressive promotional activity while lacking full network owner economics, their margins are generally lower than those of MNOs. The most recent disclosed figure was Charter's wireless EBITDA margin of approximately -11% in FY22, when the business had significantly fewer subscribers than it does today. Ultimately, MVNO profitability depends on several factors, including the degree of traffic offloading, the level of wholesale rates, and operating scale. In addition, SG&A efficiency generally improves as subscriber bases grow. As the two largest MVNOs in the U.S., Comcast and Charter have both achieved positive EBITDA in their mobile businesses as scale has increased ([Exhibit 11](#)).

Starlink's ability to offload MVNO traffic onto its own satellite network in away-from-home use cases could produce meaningful cost savings under an MVNO model. For example, increasing traffic offloading from approximately 88%, which is similar to current cable MVNO levels, to 91% would reduce wholesale MVNO fees by roughly 25%, all else being equal ([Exhibit 12](#)).

Relatively small increases in offloading can therefore generate substantial savings. These savings could translate into higher MVNO margins for Starlink, support higher wholesale rates charged by MNO partners, and/or enable lower prices for end customers ([Exhibit 13](#)).

An MVNO Strategy May Make Sense for Starlink if It Expands the Broadband Opportunity

The cash flow profile of a Starlink MVNO customer could vary significantly depending on the go-to-market strategy and customer adoption patterns. Device subsidies, promotional credits, and handset installment plans typically represent a large portion of customer acquisition costs. For example, assume Starlink acquires a new mobile account with two lines, and both subscribers purchase a new \$1,100 smartphone. If Starlink offers a trade-in credit equal to approximately 75% of the new device value (\$825 per device), while the traded-in devices can only be resold for roughly 60% of the new device value (\$660), the transaction would create an immediate initial customer acquisition outflow of \$880 $((\$1,100 - \$660) \times 2)$ and a device-related loss of approximately \$330 $((\$825 - \$660) \times 2)$. In addition, financing the new devices over a 36-month installment period would create an additional working-capital burden and time-value-of-money cost.

Assuming Starlink prices mobile service at approximately \$35 per line per month, similar to Comcast and Charter, and achieves wireless EBITDA margins of roughly 25% due to greater traffic offloading than cable MVNOs, the company could generate approximately \$650 of cumulative profit over the first three years of the customer relationship after accounting for pricing increases, account growth, and margin expansion.

The overall economics of an MVNO strategy also depend on broadband bundling dynamics. Some mobility customers may not subscribe to Starlink broadband services, while others may bundle broadband and wireless in exchange for promotional discounts. For bundled customers, two outcomes are possible. First, the customer may already be a Starlink broadband subscriber, in which case the primary benefit is likely lower broadband churn, partially offset by the bundled discount. Second, the customer may be entirely new to the broadband business, creating incremental service revenue that otherwise would not have existed (potentially expanding the TAM). The proportion of customers that fall into each category is a critical determinant of the overall economic value of the bundle (and whether the net bundle economics are either positive or negative).

Importantly, the customer relationship does not necessarily end once the initial 36-month handset financing cycle is complete. The upgrade and replacement cycle may repeat, creating another multiyear stream of customer cash flows and acquisition costs ([Exhibit 14](#)).

As discussed above, the internal rate of return (IRR) of a Starlink MVNO strategy depends on several variables. Two of the most important are: (1) the trade-in credit offered as a percentage of new device value and (2) among customers who bundle broadband and wireless services, the percentage that are new broadband subscribers rather than existing Starlink broadband customers. We assume a 40% broadband bundle rate, which is broadly consistent with AT&T's fiber and wireless bundling penetration.

Under the assumptions presented in [Exhibit 15](#), the IRR approaches 0% when the trade-in value offered is equal to the full value of the new device, effectively eliminating future customer device payments. If trade-in credits exceed the value of the new devices through additional incentives, such as prepaid cards or promotional credits, the IRR becomes negative. Conversely, when trade-in values are lower than the price of new devices, the economics improve and the IRR turns positive. A higher proportion of new broadband customers within bundled accounts can further enhance returns. Under the right combination of customer acquisition costs, traffic offloading benefits, and broadband attachment rates, a Starlink MVNO strategy could generate returns above its cost of capital.

Owning the customer relationship may matter more than maximizing wireless margins

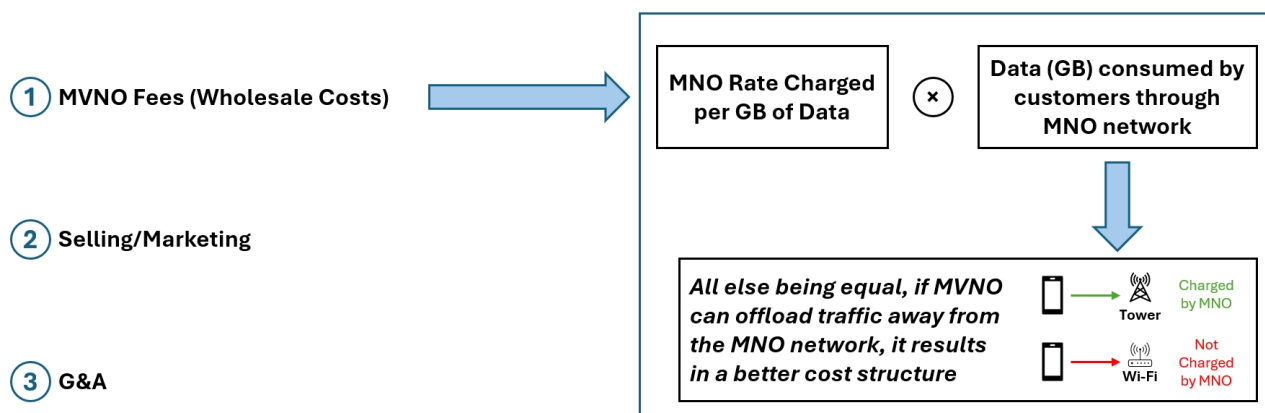
There are at least three potential objectives for SpaceX in developing a wireless business. The first is the growth opportunity that a successful standalone wireless service could provide at scale. Achieving this would require Starlink to capture a meaningful share of the wireless market. Starlink's MVNO economics could be materially stronger than those of a typical MVNO due to D2D offloading, potentially supporting this objective.

The second objective is to expand Starlink's broadband addressable market. In many markets, connectivity is already, or is increasingly becoming, a converged product, with broadband and mobility services sold together. Offering broadband-only services can limit the total addressable market if a converged offering is unavailable. An MVNO structure would enable Starlink to bundle wireless and broadband services, increasing both customer penetration and retention in broadband, similar to the strategy employed by U.S. cable operators today.

However, the larger strategic opportunity may extend beyond wireless economics altogether. In a future where AI becomes ubiquitous and the smartphone remains the primary edge device, owning the customer relationship may matter more than maximizing wireless margins. Viewed through this lens, the objective is not necessarily to optimize the economics of a standalone wireless business, whether through an MVNO arrangement or a facilities-based network. Rather, the goal may be to achieve scale, establish distribution, and build direct customer relationships as quickly as possible. If that is the objective, an MVNO-based hybrid network could represent the most practical path forward ([Exhibit 16](#)).

EXHIBIT 9: MVNO's profitability is dependent on its ability to offload traffic away from the MNO network

Elements of MVNO's Cost to Serve



Source: Bernstein estimates and analysis

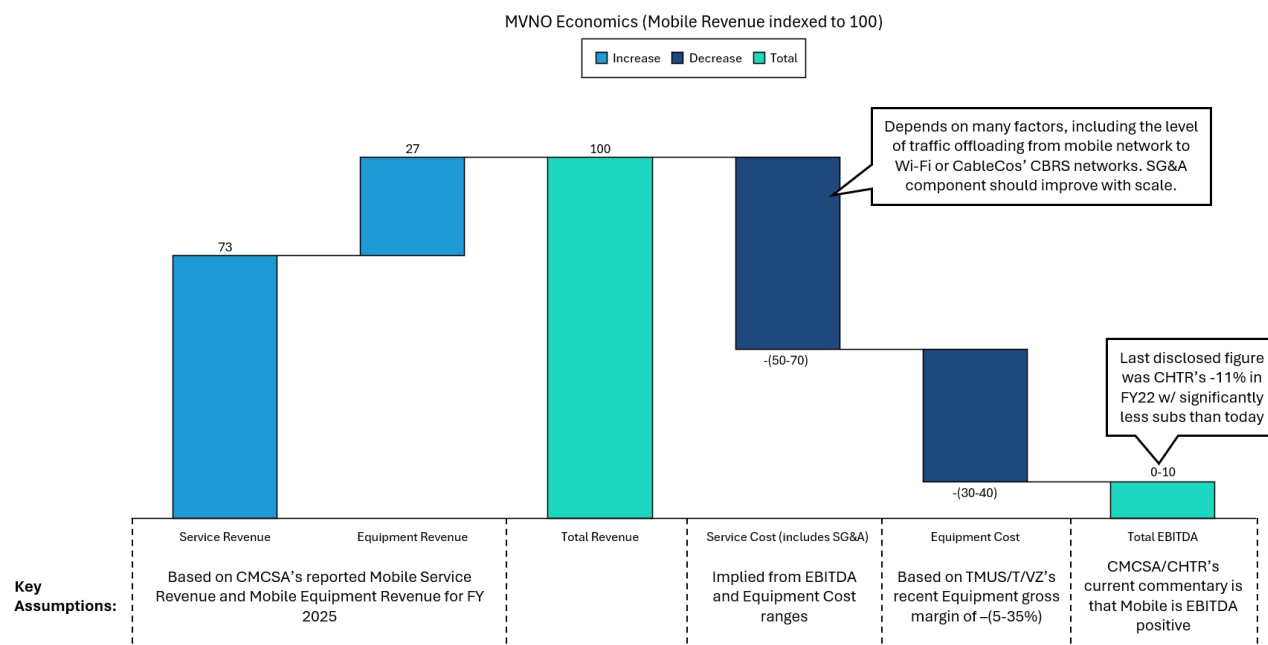
EXHIBIT 10: A potential Starlink MVNO model could have an advantage over CableCo's in that it may be able to offload significant traffic to its satellite constellation and - perhaps - femtocells

	Potential Differences in MVNO Offloading	
	Cable MVNOs	Potential Starlink MVNO
Normal US Handset Offloading to Wi-Fi	~80-90% of handset data traffic is offloaded to Wi-Fi due to consumption patterns in the US	
Wi-Fi Hotspots & CBRS Spectrum Terrestrial Network	CHTR & CMCSA achieve ~87-90% offloading to their extensive Wi-Fi and CBRS networks (+2-7% vs. MNOs)	No owned terrestrial network
Starlink Mobile Satellite Constellation/Femtocell	No owned satellite network	Starlink could potentially offload traffic to its satellite network or femtocells for away-from-home use cases

Unlike CableCo's CBRS network, the satellite constellation & femtocell combination is ubiquitous, which could result in higher offloading

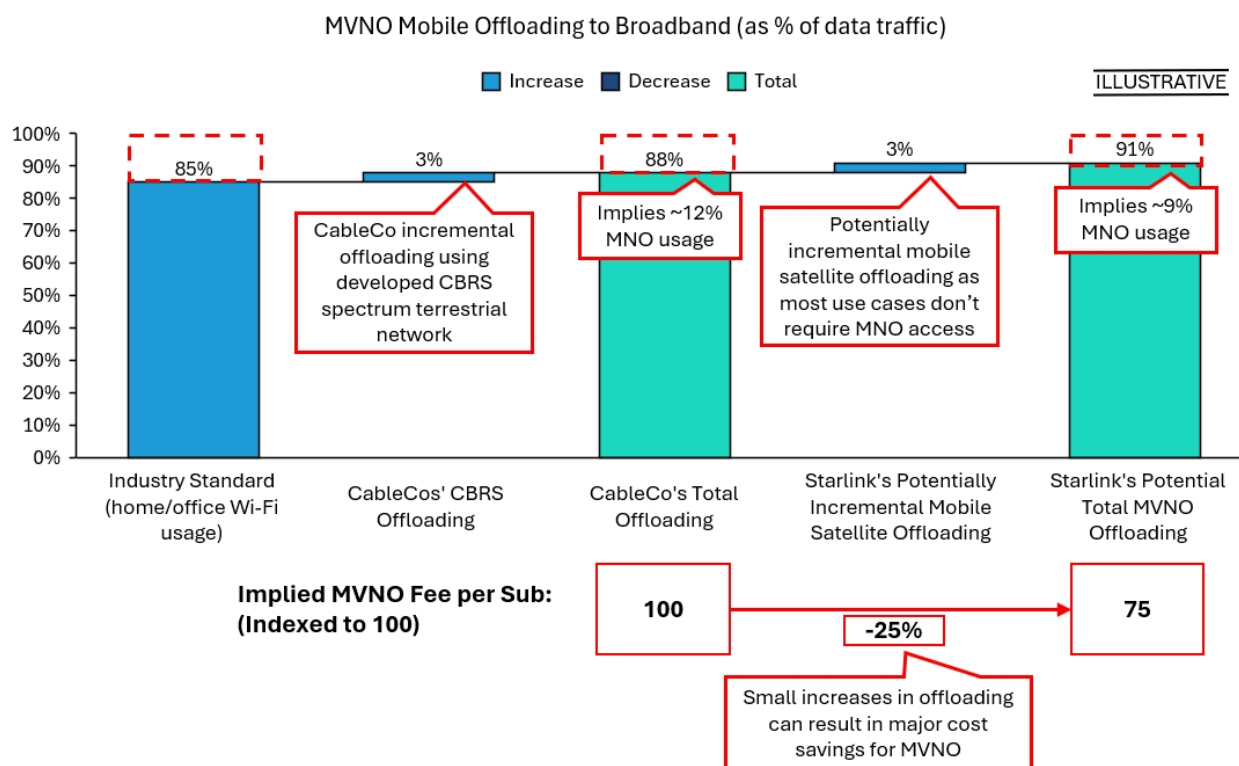
Source: OpenSignal, Company filings, Bernstein estimates and analysis

EXHIBIT 11: CMCSA and CHTR have achieved positive EBITDA on their Mobile businesses as the largest MVNOs in the US



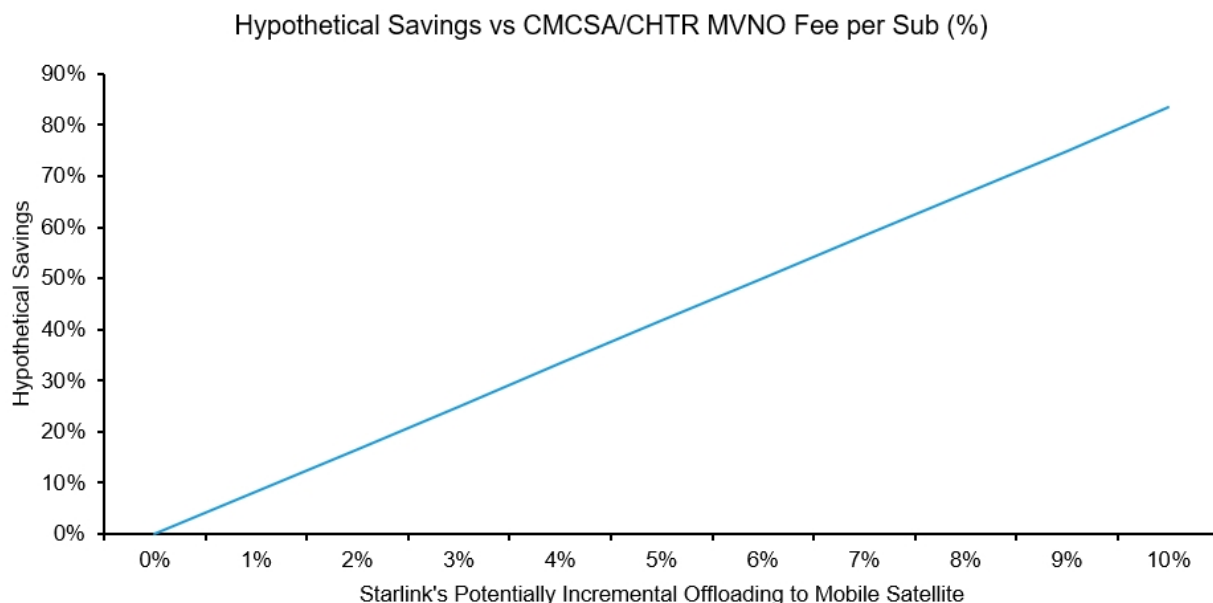
Source: Company filings, Bernstein estimates and analysis

EXHIBIT 12: Starlink's capacity to offload MVNO traffic to its own mobile satellite network in away-from-home use cases could result in major cost savings in an MVNO model



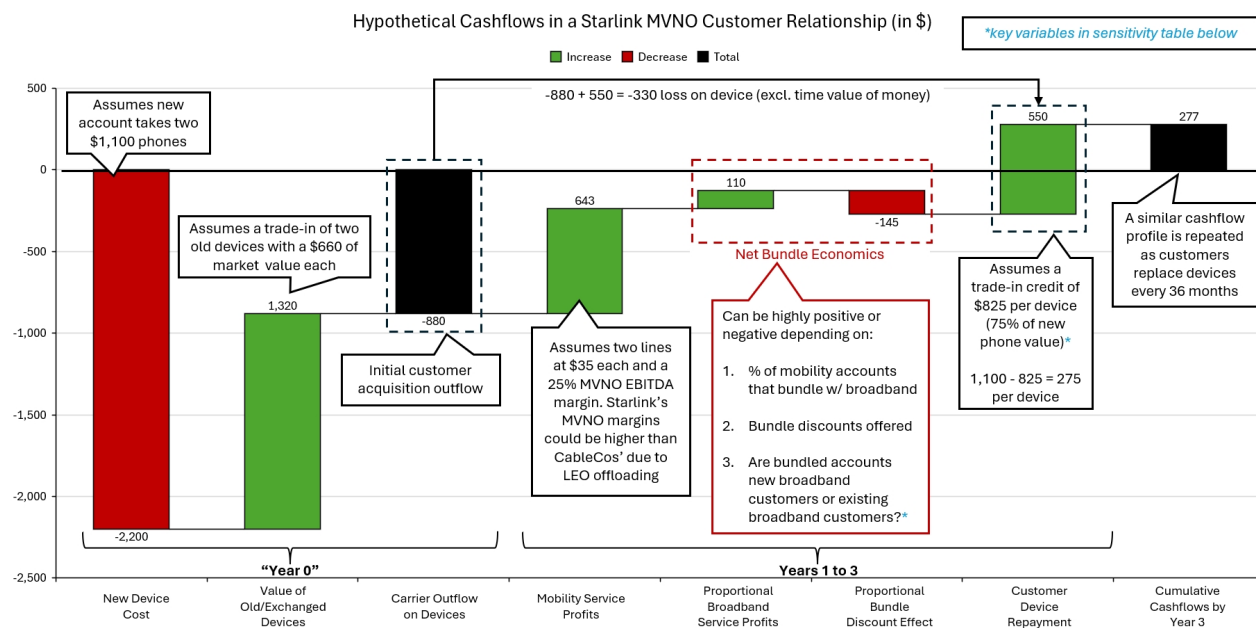
Source: OpenSignal, Company filings, Bernstein estimates and analysis

EXHIBIT 13: Small offloading increments can result in major cost savings for Starlink. This can result in better MVNO margins for Starlink, higher fees per bit charged by MNOs, and/or lower prices for Starlink Mobile customers

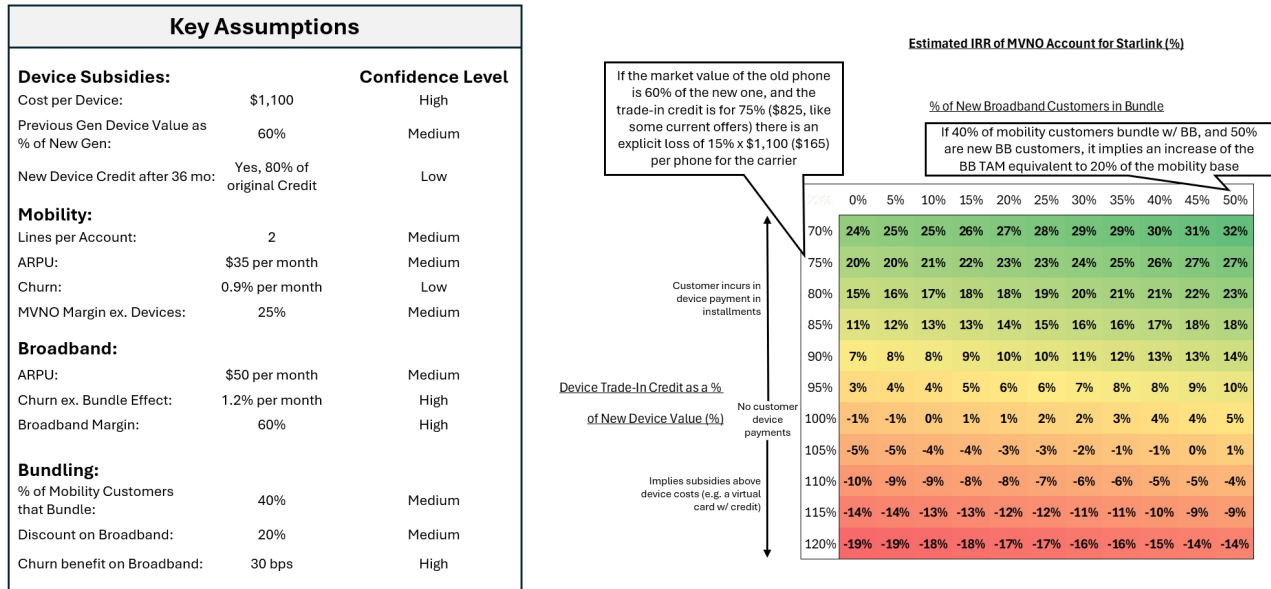


Source: Company filings, Bernstein estimates and analysis

EXHIBIT 14: The cashflow profile of a Starlink MVNO customer relationship may look very different depending on strategy and market reception. One key variable is by what degree do bundles increase the TAM for broadband customers



Source: Company websites, Bernstein estimates and analysis

EXHIBIT 15: **The Starlink wireless business could expand its broadband TAM and produce better returns****Analysis of Estimated IRR of MVNO Account for Starlink**

Source: Company websites, Company filings, Bernstein estimates and analysis

EXHIBIT 16: **The more important strategic opportunity for a Starlink MVNO may extend beyond wireless economics altogether. If AI becomes ubiquitous, owning the customer relationship may matter more than maximizing wireless margins****Potential objectives of Wireless for SpaceX**

Wireless growth	- Standalone merits of wireless at scale can be attractive - Would need to believe Starlink can capture a meaningful share of the market
Expand Starlink's BB addressable market	- Many markets, including the US, is already or increasingly converged - BB-only play limits TAM
Own the customer for future AI-play	- If/when AI is ubiquitous, owning the ultimate edge device can be a strategic advantage to monetize - SpaceX's AI can be an embedded service for Starlink Mobile subs

Does MVNO structure support?

☒	- Standalone MVNO economics is unattractive and challenging to scale - However, economics could be materially better than typical MVNOs with D2D offloading
☒	Wireless-BB bundled offers can increase Starlink BB penetration and retention – works even with low MVNO margins (like US CableCos)
☒	Starlink would own the customer relationship (not MNO providing network as a service to Starlink)

- Can Starlink-branded MVNO bundled with Starlink BB achieve meaningful scale to achieve its objectives?
- Possibly more than traditional MVNOs

Source: Bernstein analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 17: SpaceX Annual Income Statement

SpaceX Annual Income Statement (\$m)								
Fiscal Year Ending December 31	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Space	3,796	4,086	4,794	7,560	7,513	7,224	7,930	9,167
Connectivity	7,599	11,387	17,037	27,727	53,120	94,679	140,998	205,908
AI	2,620	3,201	24,571	115,088	195,835	229,164	305,581	439,753
Total Revenues	\$ 14,015	\$ 18,674	\$ 46,402	\$ 150,375	\$ 256,469	\$ 331,067	\$ 454,509	\$ 654,827
Cost of revenue	(7,996)	(9,451)	(17,167)	(46,740)	(80,978)	(101,693)	(124,050)	(178,953)
Research & Development	(3,464)	(8,643)	(23,201)	(67,204)	(49,285)	(61,761)	(77,515)	(110,456)
Selling, General, and administrative	(1,813)	(2,644)	(4,309)	(8,742)	(13,599)	(16,750)	(22,213)	(29,195)
Restructuring charges	(213)	(487)	9	-	-	-	-	-
Impairment	(63)	(38)	-	-	-	-	-	-
Total cost and expenses	-13,549	-21,263	-44,669	-122,687	-143,862	-180,204	-223,778	-318,604
Space	21	(657)	(2,047)	1,262	896	236	503	874
Connectivity	2,006	4,423	6,082	12,006	24,626	44,437	67,304	102,316
AI	(1,561)	(6,355)	(2,302)	14,419	87,084	106,190	162,924	233,033
Operating income	\$ 466	\$ (2,589)	\$ 1,734	\$ 27,688	\$ 112,607	\$ 150,863	\$ 230,731	\$ 336,223
Interest income (expense)	(1,209)	(1,453)	(2,129)	(11,511)	(16,320)	(22,010)	(20,989)	(20,796)
Other income (expense), net	985	-177	-1962	0	0	0	0	0
Income (loss) before income taxes	\$ 242	\$ (4,219)	\$ (2,357)	\$ 16,177	\$ 96,287	\$ 128,853	\$ 209,741	\$ 315,427
Provisions for (benefit from) income taxes	549	(718)	(82)	(99)	14,202	(2,414)	(4,828)	(39,563)
Net income	\$ 791	\$ (4,937)	\$ (2,439)	\$ 16,078	\$ 110,489	\$ 126,439	\$ 204,913	\$ 275,864
Shares outstanding (mn)								
Basic			11,789	11,789	11,789	11,789	11,789	11,789
Diluted			12,344	12,344	12,344	12,344	12,344	12,344
EPS (\$)								
Basic EPS			-0.21	1.36	9.37	10.73	17.38	23.40
Diluted EPS			-0.20	1.30	8.95	10.24	16.60	22.35
Adjusted EBITDA	\$ 5,350	\$ 6,584	\$ 21,206	\$ 85,583	\$ 180,205	\$ 233,754	\$ 339,093	\$ 486,986
Space	1,154	653	-561	2,817	3,122	2,939	3,768	4,556
Connectivity	3,849	7,168	10,520	17,075	30,947	53,498	78,205	114,737
AI	347	-1,237	11,247	65,690	146,136	177,318	257,120	367,694

Source: Company report, Bernstein analysis and estimates

EXHIBIT 18: **SpaceX Quarterly Income Statement**

SpaceX Quarterly Income Statement (\$m)											
Fiscal Year Ending December 31											
	2024					2025					2026E
	12 mo	Q1	Q2	Q3	Q4	12 mo	Q1	Q2	Q3	Q4	12 mo
Space	3,796	865	746	1,666	809	4,086	619	962	1,538	1,675	4,794
Connectivity	7,599	2,475	2,588	3,096	3,228	11,387	3,257	4,291	4,448	5,040	17,037
AI	2,620	727	737	814	923	3,201	818	2,561	6,713	14,479	24,571
Total Revenues	\$ 14,015	\$ 4,067	\$ 4,071	\$ 5,576	\$ 4,960	\$ 18,674	\$ 4,694	\$ 7,814	\$ 12,699	\$ 21,195	\$ 46,402
Cost of revenue	(7,996)	(1,962)	(2,282)	(2,706)	(2,501)	(9,451)	(2,388)	(3,495)	(4,591)	(6,693)	(17,167)
Research & Development	(3,464)	(1,557)	(1,958)	(2,291)	(2,837)	(8,643)	(3,514)	(3,548)	(5,490)	(10,649)	(23,201)
Selling, General, and administrative	(1,813)	(493)	(606)	(575)	(970)	(2,644)	(746)	(912)	(1,091)	(1,561)	(4,309)
Restructuring charges	(213)	(4)	(189)	(278)	(16)	(487)	11	(2)	-	-	9
Impairment	(63)	(24)	(6)	1	(9)	(38)	-	-	-	-	-
Total cost and expenses	-13,549	-4,040	-5,041	-5,849	-6,333	-21,263	-6,637	-7,957	-11,172	-18,903	-44,669
Operating income (loss) from operations	\$ 466	\$ 27	\$ (970)	\$ (273)	\$ (1,373)	\$ (2,589)	\$ (1,943)	\$ (143)	\$ 1,527	\$ 2,292	\$ 1,734
Interest income (expense)	(1,209)	(330)	(312)	(359)	(452)	(1,453)	(451)	(289)	(694)	(694)	(2,129)
Other income (expense), net	985	(211)	413	140	(519)	(177)	(1,876)	(86)	-	-	(1,962)
Income (loss) before income taxes	\$ 242	\$ (514)	\$ (869)	\$ (492)	\$ (2,344)	\$ (4,219)	\$ (4,270)	\$ (518)	\$ 833	\$ 1,598	\$ (2,357)
Provisions for (benefit from) income taxes	549	(14)	(138)	(562)	(4)	(718)	(6)	(23)	-	(53)	(82)
Net income	\$ 791	\$ (528)	\$ (1,007)	\$ (1,054)	\$ (2,348)	\$ (4,937)	\$ (4,276)	\$ (541)	\$ 833	\$ 1,544	\$ (2,439)
Shares outstanding (mn)											
Basic							4,496	5,864	11,789	11,789	11,789
Diluted							5,068	5,864	12,344	12,344	12,344
EPS (\$)											
Basic EPS							-\$0.95	-\$0.09	\$0.07	\$0.13	-\$0.21
Diluted EPS							-\$0.84	-\$0.09	\$0.07	\$0.13	-\$0.20
Adjusted EBITDA	5,350	1,730	1,212	2,168	1,474	6,584	1,127	3,538	6,800	9,741	21,206
Space	1,154	224	-94	672	-149	653	-351	-205	80	-85	-561
Connectivity	3,849	1,618	1,583	1,921	2,046	7,168	2,087	2,597	2,799	3,037	10,520
AI	347	-112	-277	-425	-423	-1,237	-609	1,146	3,921	6,789	11,247

Source: Company report, Bernstein analysis and estimates

EXHIBIT 19: **SpaceX Annual Balance Sheet**

SpaceX Annual Balance Sheet (\$m)								
Fiscal Year Ending December 31	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Cash and cash equivalents	11,385	24,747	63,003	59,295	128,410	145,412	236,854	225,752
Marketable securities	800	0	6,487	6,487	6,487	6,487	6,487	6,487
Account receivables (net allowance credit loss)	1,052	1,579	3,941	8,240	18,761	27,483	39,100	40,545
Inventory	2,003	2,416	4,341	7,504	5,680	4,764	4,724	3,824
Prepaid expenses and other current assets	868	2,210	2,550	4,035	3,234	3,040	3,265	2,777
Total current assets	16,108	30,952	80,322	85,560	162,572	187,186	290,431	279,386
Property, plant, and equipment, net	21,147	42,602	116,519	328,948	550,165	801,617	1,074,408	1,345,960
Finance lease right-of-use assets	1,686	1,260	892	533	0	0	0	0
Intangible assets, net	2,211	1,548	0	0	0	0	0	0
Digital assets	1,749	1,637	0	0	0	0	0	0
Goodwill	11,129	11,809	0	0	0	0	0	0
Other assets	2,336	2,130	16,390	16,007	1,836	9,700	17,685	25,737
Total non current assets	40,258	60,986	133,801	345,488	552,001	811,317	1,092,093	1,371,697
Total assets	56,366	91,938	214,123	431,048	714,572	998,503	1,382,524	1,651,083
Accounts payable	4,413	11,792	8,217	18,606	21,653	28,836	37,691	41,355
Deferred revenue, current	10,179	12,116	27,099	43,007	32,469	31,782	42,724	30,449
Accrued expenses and other liabilities	1,508	2,569	8,286	12,531	9,127	8,712	9,449	8,045
Total current liabilities	16,100	26,477	43,602	74,145	63,249	69,331	89,864	79,849
Debt	12,262	21,659	38,700	209,288	408,001	550,241	699,647	693,188
Finance Lease Liabilities	1,531	1,237	752	393	0	0	0	0
Deferred Tax Liability / (Asset)	-725	-113	8	84	-14,305	-5,136	4,032	13,201
Other Long-Term Liabilities	1,394	1,353	1,293	1,293	1,293	1,293	1,293	1,293
Total non-current liabilities	14,462	24,136	40,753	211,058	394,989	546,398	704,973	707,682
Total liabilities	30,562	50,613	84,355	285,203	458,238	615,729	794,837	787,532
Convertible Preferred Stock	20,941	38,752	0	0	0	0	0	0
Additional Paid-in Capital	35,868	37,710	167,344	167,344	167,344	167,344	167,344	167,344
Retained Earnings	-31,005	-35,137	-37,576	-21,498	88,991	215,430	420,343	696,207
Total shareholders' equity	25,804	41,325	129,768	145,846	256,335	382,774	587,687	863,551
Total liabilities and equity	56,366	91,938	214,122	431,048	714,573	998,503	1,382,524	1,651,082
Total Debt	13,793	22,896	39,452	209,681	408,001	550,241	699,647	693,188
Net debt	2,408	-1,851	-23,552	150,386	279,590	404,829	462,793	467,436

Source: Company report, Bernstein analysis and estimates

EXHIBIT 20: **SpaceX Annual Cash Flow**

SpaceX Annual Cash Flow (\$m)								
Fiscal Year Ending December 31	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Net income (loss)	791	(4,937)	(2,439)	16,078	110,489	126,439	204,913	275,864
Depreciation & Amortization	3,824	6,701	16,586	54,611	63,819	78,703	103,716	145,687
Stock Based Compensation	784	1,947	2,896	3,284	3,779	4,189	4,647	5,077
Impairment & Other Charges	63	38	-	-	-	-	-	-
Non-Cash Interest Income and Other	(1,359)	947	1,433	(703)	(1,031)	(330)	(1,340)	(1,179)
Accounts Receivable	(347)	(543)	(2,362)	(4,299)	(10,521)	(8,722)	(11,617)	(1,445)
Inventory	(309)	(413)	(1,925)	(3,163)	1,825	915	40	900
Prepaid Expenses & Other	(328)	(673)	(340)	(1,485)	800	194	(225)	488
Accounts Payable	472	709	(3,575)	10,390	3,047	7,183	8,855	3,665
Deferred Revenue	1,876	1,929	14,983	15,908	(10,538)	(686)	10,941	(12,274)
Other Liabilities	309	1,080	5,717	4,245	(3,404)	(415)	737	(1,405)
Net cash provided by operating activities	\$ 5,776	\$ 6,785	\$ 30,973	\$ 94,866	\$ 158,264	\$ 207,469	\$ 320,666	\$ 415,377
Purchase of PPE	(11,163)	(20,619)	(90,502)	(267,040)	(285,036)	(330,155)	(376,507)	(417,238)
Other investing activities, net	367	1,044	(7,206)	-	-	-	-	-
Net cash used in investing activities	\$ (10,796)	\$ (19,575)	\$ (97,708)	\$ (267,040)	\$ (285,036)	\$ (330,155)	\$ (376,507)	\$ (417,238)
Debt Issuance / (Paydown)	(77)	9,131	17,041	170,588	198,712	142,241	149,406	(6,459)
Finance Lease Payments	(154)	(295)	(374)	(470)	(393)	-	-	-
Equity Issuance / (Repurchases) **	12,061	17,514	88,746	(1,653)	(2,432)	(2,553)	(2,123)	(2,781)
Net cash provided by financing activities	\$ 11,830	\$ 26,350	\$ 105,413	\$ 168,465	\$ 195,887	\$ 139,687	\$ 147,284	\$ (9,241)
Effect of exchange rate changes on cash and cash equivalents	1	63	(42)	-	-	-	-	-
Net change in cash and cash equivalents and restricted cash	\$ 6,811	\$ 13,623	\$ 38,635	\$ (3,708)	\$ 69,115	\$ 17,002	\$ 91,443	\$ (11,102)
Cash and cash equivalents and restricted cash, beginning year	\$ 4,690	\$ 11,501	\$ 25,124	\$ 63,759	\$ 60,051	\$ 129,166	\$ 146,168	\$ 237,610
Cash and cash equivalents and restricted cash, end year	\$ 11,501	\$ 25,124	\$ 63,759	\$ 60,051	\$ 129,166	\$ 146,168	\$ 237,610	\$ 226,508
FCF	(5,387)	(13,834)	(59,530)	(172,174)	(126,772)	(122,685)	(55,841)	(1,861)
Conversion to net income %	-681%	280%	2440%	-1071%	-115%	-97%	-27%	-1%
Working Capital	(12,177)	(20,272)	(32,770)	(54,366)	(35,575)	(34,044)	(42,774)	(32,703)

Source: Company report, Bernstein analysis and estimates

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY

This research publication covers six or more companies. For valuation methodology and other company disclosures:

Please visit: <https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action>.

Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RISKS

This research publication covers six or more companies. For risks and other company disclosures:

Please visit: <https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action>.

Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

Prior to April 1, 2024, Sanford C. Bernstein & Co., LLC. issued the ratings and price target information in the graph(s) below for the following companies: Comcast Corp and Charter Communications Inc.

PRICE CHARTS/ RATINGS AND PRICE TARGET HISTORY

This research publication covers six or more companies. For price chart and other company disclosures, please visit <https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action> or you can write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

CONFLICTS OF INTEREST

All statements in this report attributable to Gartner represent Bernstein's interpretation of data, research opinion or viewpoints published as part of a syndicated subscription service by Gartner, Inc., and have not been reviewed by Gartner. Each Gartner publication speaks as of its original publication date (and not as of the date of this report). The opinions expressed in Gartner publications are not representations of fact, and are subject to change without notice.

Stacy A. Rasgon maintains long positions in various crypto currencies.

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Space Exploration Technologies Corporation, AT&T Inc, T-Mobile US Inc and Comcast Corp.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Space Exploration Technologies Corporation, AT&T Inc, Verizon Communications Inc, T-Mobile US Inc, Comcast Corp and American Tower Corporation.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Verizon Communications Inc, T-Mobile US Inc and Comcast Corp.

Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Comcast Corp.

Bernstein and/or affiliates expect to receive or intend to seek compensation for investment banking services in the next three months from Space Exploration Technologies Corporation and AT&T Inc.

Affiliates of Bernstein managed or co-managed in the past twelve months a public offering of securities of Space Exploration Technologies Corporation, AT&T Inc, T-Mobile US Inc and Comcast Corp.

Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Space Exploration Technologies Corporation, AT&T Inc, T-Mobile US Inc and Comcast Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Space Exploration Technologies Corporation, AT&T Inc, Verizon Communications Inc, T-Mobile US Inc, Comcast Corp, Charter Communications Inc, American Tower Corporation, Crown Castle Inc and SBA Communications Corporation.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: AT&T Inc, Verizon Communications Inc, T-Mobile US Inc, Comcast Corp, Charter Communications Inc and American Tower Corporation.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Global Solution Center Morocco, Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore
- +81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject

securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations,

etc.”) of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as “relevant persons”). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore (“SFA”). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People’s Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People’s Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the “PRC”) in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. (“Daiwa”), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;

- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
 - do not have any financial interest in the subject company
 - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
 - is not engaged in any investment banking activities for Indian companies, as such;
 - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
 - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
 - have not received compensation for brokerage services from the subject company in the past twelve months;
 - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
 - do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
 - do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](http://Sanford C. Bernstein (India) Private Limited) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read

all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.